

Reaction kinetics and pattern formation in a diffusive epidemic model with an infection-dependent recovery rate

Wael El Khateeb, Chanaka Kottegoda, Chunhua Shan

JMM Contributed Talk Session on Mathematical Biology



Sections:

- 1) Model and Mechanisms
- 2) ODE Dynamics
- 3) PDE Patterns

Location: Lombardy region, Italy
Disease: SARS-CoV-2 (COVID-19)

Events in chronological order:

Feb 20: First COVID-19 case reported in Italy.

Feb 23: Ten municipalities declared a “red zone.”

Mar 4: All schools are closed nationwide.

Mar 8: Fourteen northern provinces are placed under lockdown.

Mar 10: The lockdown is extended to all of Italy.

Mar 15: ICU capacity in Lombardy reaches full occupancy.



By late March 2020, Lombardy was Italy's COVID-19 epicenter.

Regional and national authorities moved quickly to expand ICU capacity by centralizing triage, transferring patients to less crowded hospitals, and opening the Fiera Milano field hospital.

Because Lombardy is tightly connected to nearby provinces through daily commuting and freight corridors, early transmission there helped seed outbreaks in adjacent areas.

Section 1: Model and Mechanism



Contains content © 2002 Microsoft Corp. All rights reserved.

W. El Khateeb, C. Kottegoda, C. Shan

Relief came from targeted actions that directly addressed the bottlenecks:

- 1) Increase ICU capacity and boost on-site care capabilities in the worst-hit zones.
- 2) Trim movement at hotspot nodes through transfers, staggered shifts, and reduced crowding on key routes.

Question: How can we include both the number of hospital beds and spatial spread in a mathematical model?

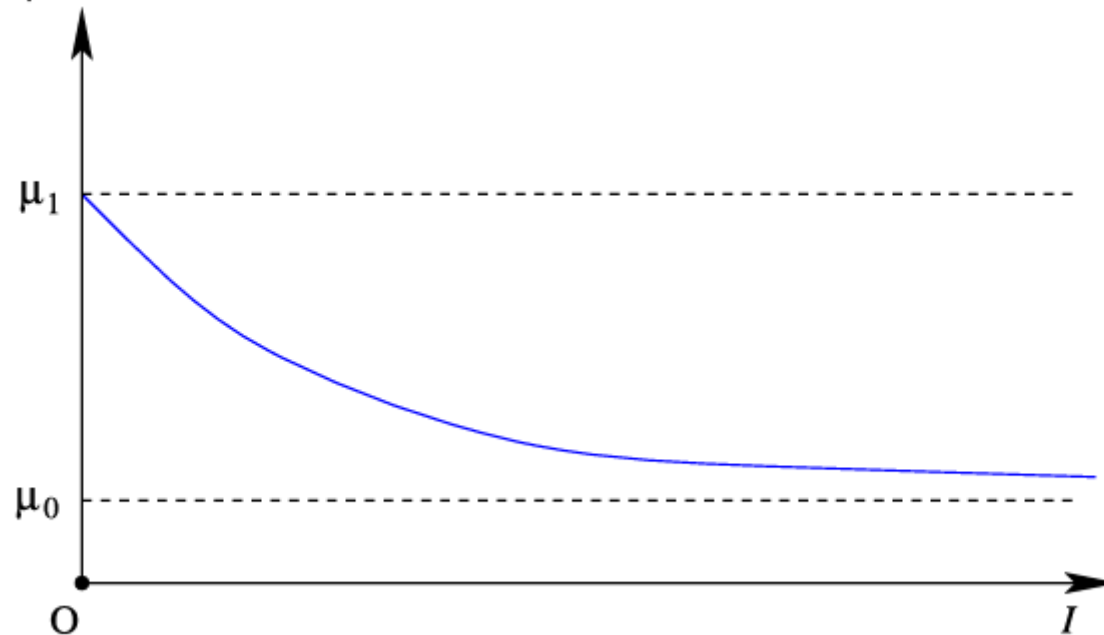
We start with a basic SIR model:

$$\begin{aligned}\frac{dS}{dt} &= A - dS - \beta SI, \\ \frac{dI}{dt} &= \beta SI - (d + \gamma)I, \\ \frac{dR}{dt} &= \gamma I - dR, \\ S(0) &= S_0, \quad I(0) = I_0, \quad R(0) = R_0.\end{aligned}\tag{1}$$

1) Hospital Bed Availability: we introduce the function:

$$h(b, I) = \mu_0 + (\mu_1 - \mu_0) \frac{b}{I + b} \quad (2)$$

Where h is the impact of hospital resource on the recovery rate, μ_0 and μ_1 are the minimum and maximum recovery rate per capita, b is the number of hospital beds and I is the number of infected individuals.



2) Spatial component: Fick's laws

Law 1 (flux): Flux moves from high to low concentration and is proportional to the gradient

$$\mathbf{J}(x, t) = -D \nabla C(x, t).$$

Law 2 (conservation + Law 1):

$$\frac{\partial C}{\partial t} = D \Delta C.$$

If there are local sources or sinks $f(C, x, t)$, conservation gives the **reaction–diffusion** form:

$$\frac{\partial C}{\partial t} = D \Delta C + f(C, x, t).$$

Under the assumption of random non-biased moving of individuals, we arrive at the final model:

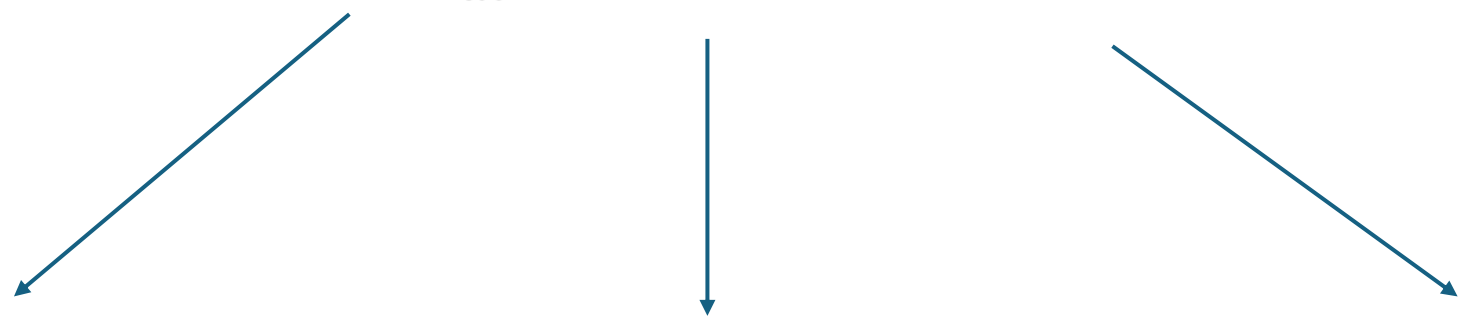
$$\begin{cases} S_t - r_1 \Delta S = A - dS - \beta SI, & x \in \Omega, t > 0; \\ I_t - r_2 \Delta I = \beta SI - dI - h(I)I, & x \in \Omega, t > 0; \\ R_t - r_3 \Delta R = h(I)I - dR, & x \in \Omega, t > 0; \\ S(x, 0) = S_0(x), I(x, 0) = I_0(x), R(x, 0) = R_0(x), & x \in \Omega; \\ \frac{\partial S}{\partial \nu} = 0, \frac{\partial I}{\partial \nu} = 0, \frac{\partial R}{\partial \nu} = 0. & x \in \partial\Omega, t > 0, \end{cases} \quad (3)$$

The recovered class is assumed to have immunity over time, and that is why it does not return to the susceptible class, and the system can be decoupled as follows:

$$\begin{cases} S_t - r_1 \Delta S = A - dS - \beta SI, & x \in \Omega, t > 0; \\ I_t - r_2 \Delta I = \beta SI - dI - h(I)I, & x \in \Omega, t > 0; \\ S(x, 0) = S_0(x) \geq 0, I(x, 0) = I_0(x) \geq 0, & x \in \Omega; \\ \frac{\partial S}{\partial \nu} = 0, \frac{\partial I}{\partial \nu} = 0. & x \in \partial\Omega, t > 0. \end{cases} \quad (4)$$

Basic Analysis: ODE model

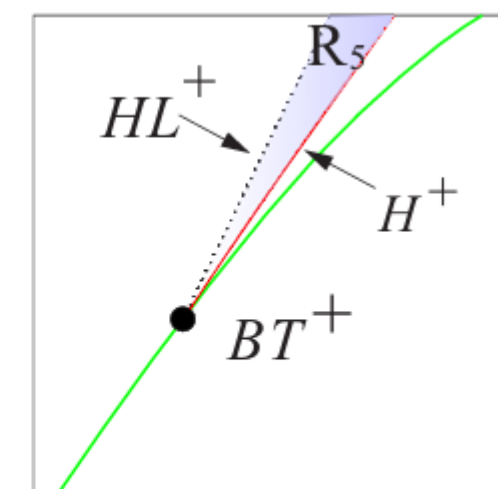
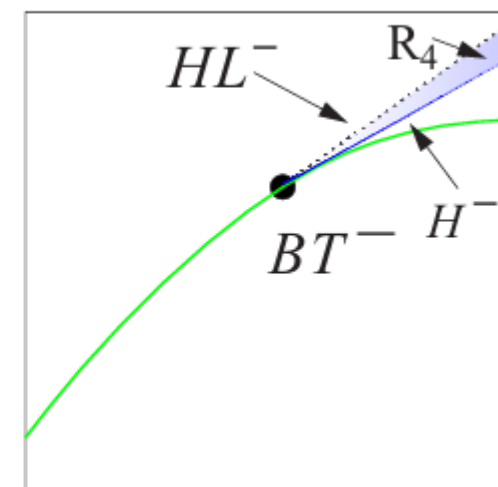
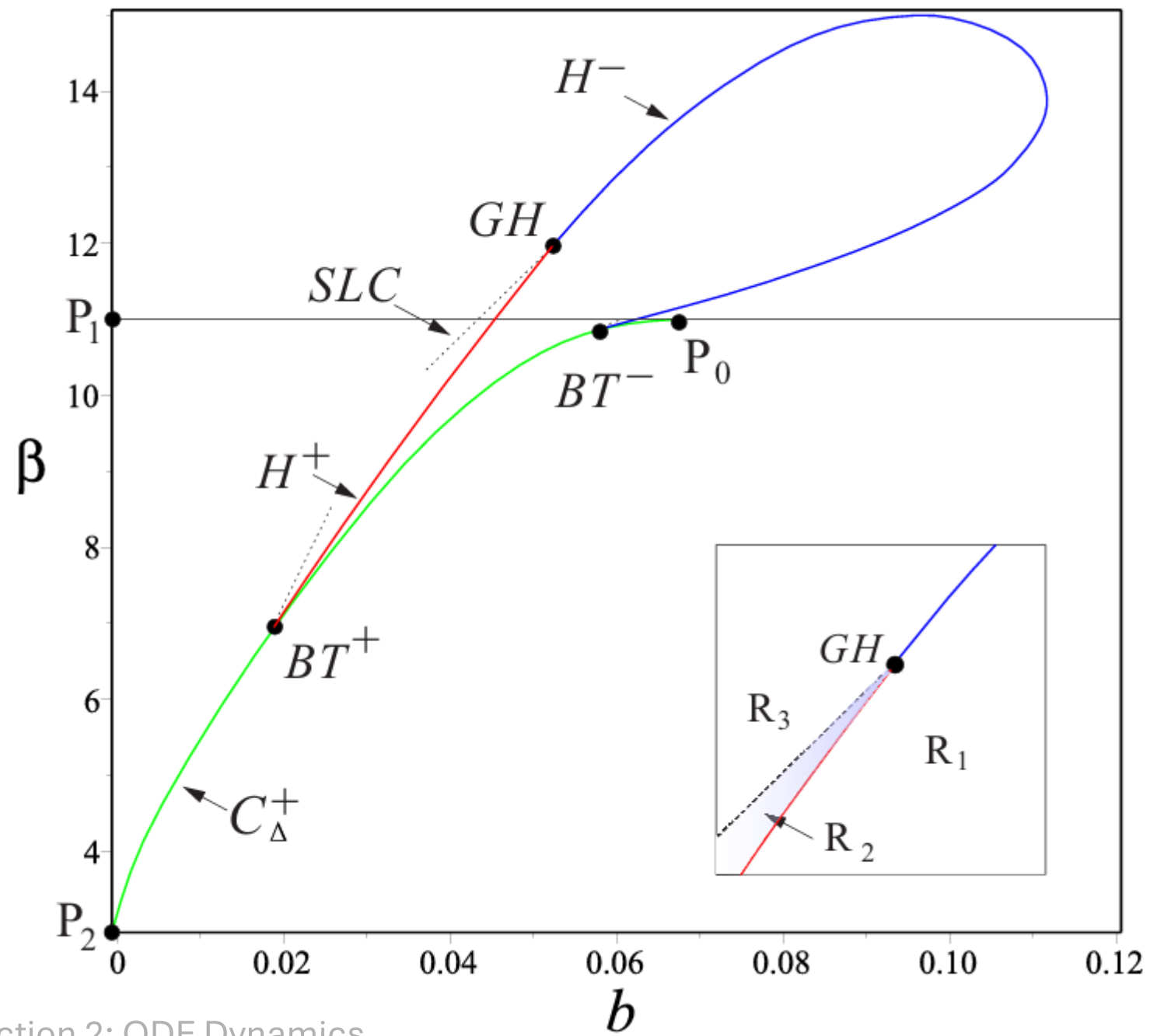
$$\begin{cases} \frac{dS}{dt} = A - dS - \beta SI, \\ \frac{dI}{dt} = \beta SI - dI - h(I)I. \end{cases} \quad (5)$$

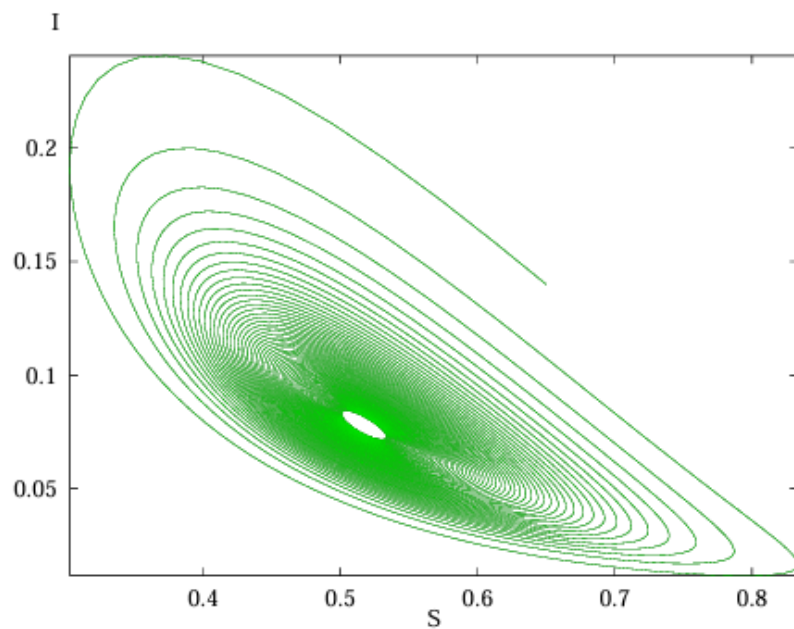


Disease Free equilibrium
 $E_0 = \left(\frac{A}{d}, 0\right)$

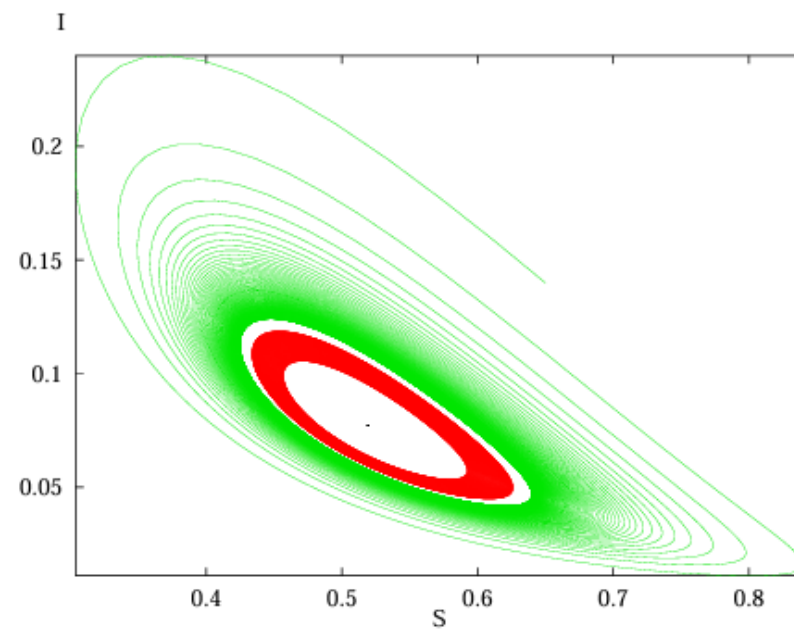
Endemic equilibrium
 $E_1 = (S_1, I_1)$

Endemic equilibrium
 $E_2 = (S_2, I_2)$

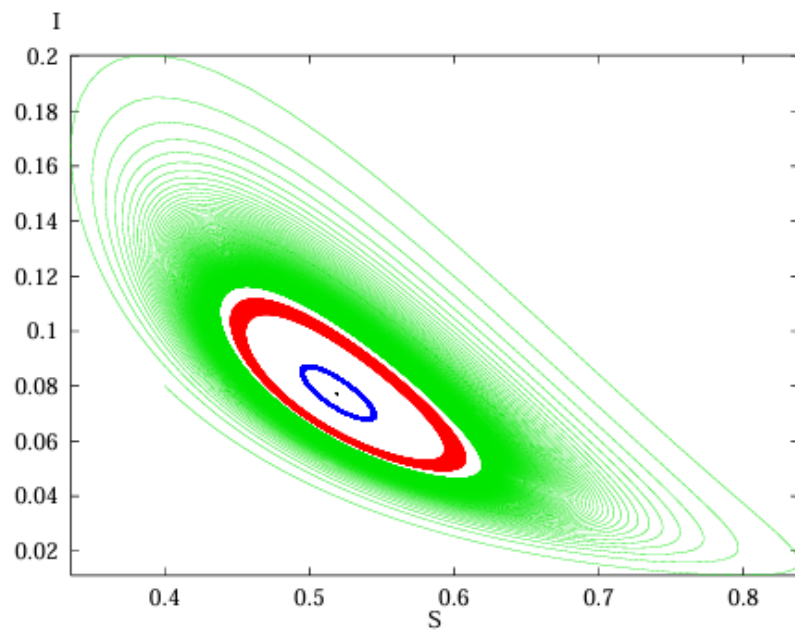




(a) E_2 is stable and no periodic solution in R_3



(b) One periodic solution in R_1



(c) Two periodic solutions in R_2

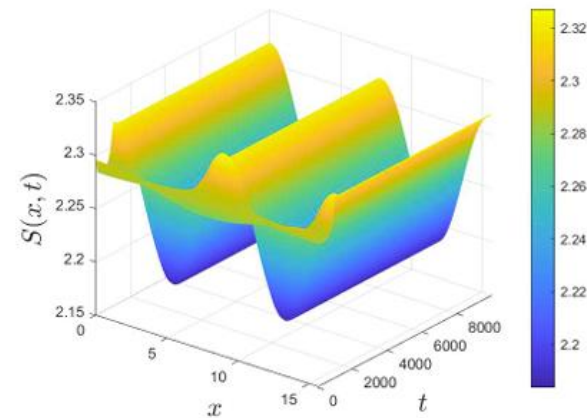
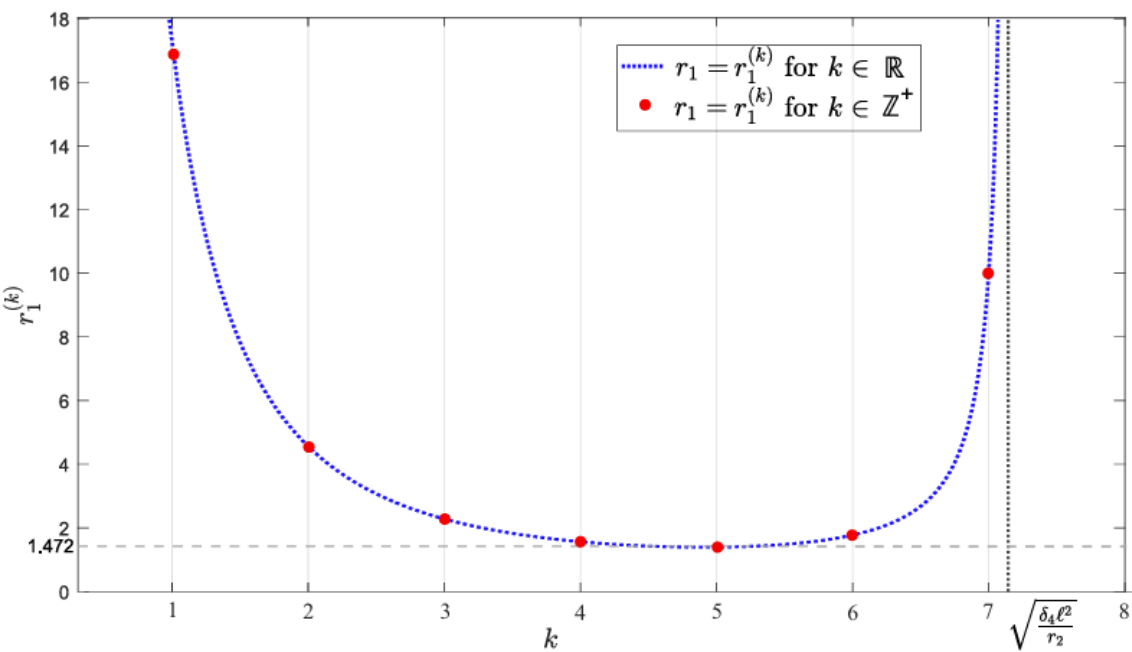
What happens when extending our system to the PDE case? Instead of 2 eigenvalues, we now have infinitely many!

$$J_k(\bar{S}, \bar{I}) = J_0(\bar{S}, \bar{I}) - \frac{k^2}{\ell^2} \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix}, \quad k \in \mathbb{Z}^+ \cup \{0\}.$$

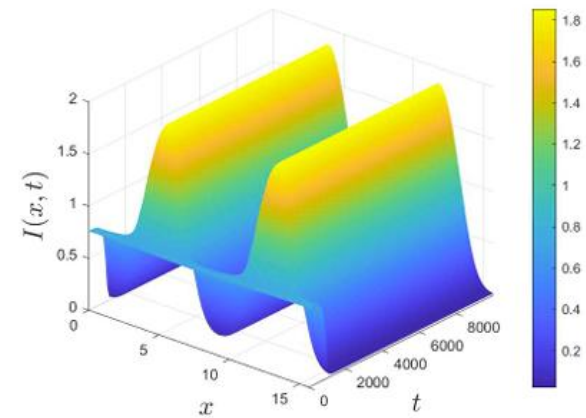
(10)

What happens when determinant equals to zero? What happens when trace vanishes? What if both disappear simultaneously?

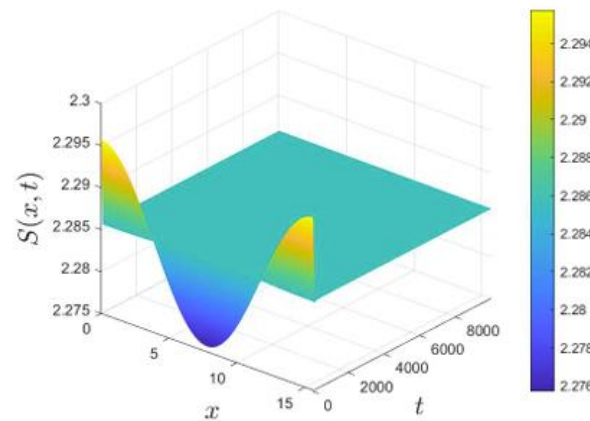
(k_1) – Turing Bifurcation



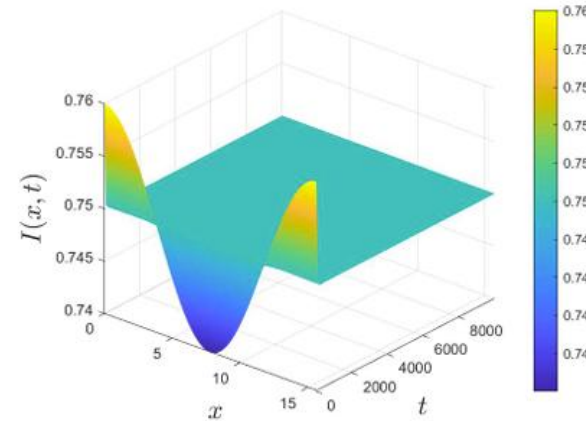
(a) Surface $S(x, t)$



(b) Surface $I(x, t)$

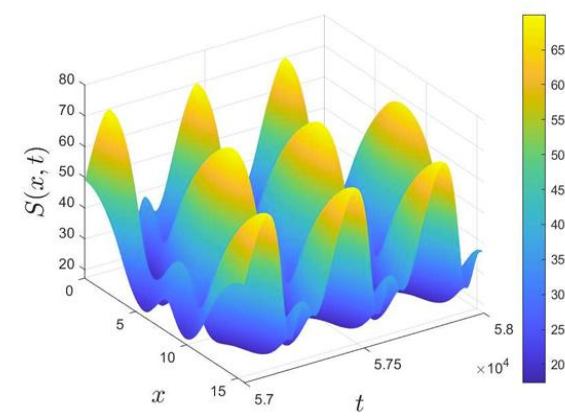
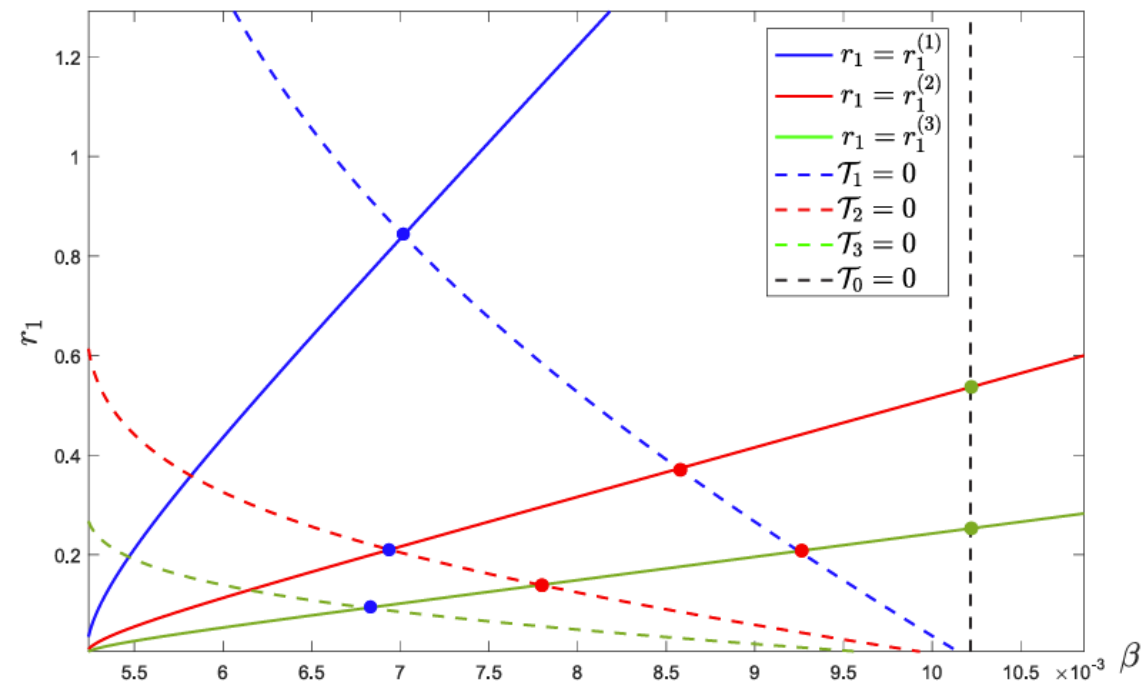


(a) Surface $S(x, t)$

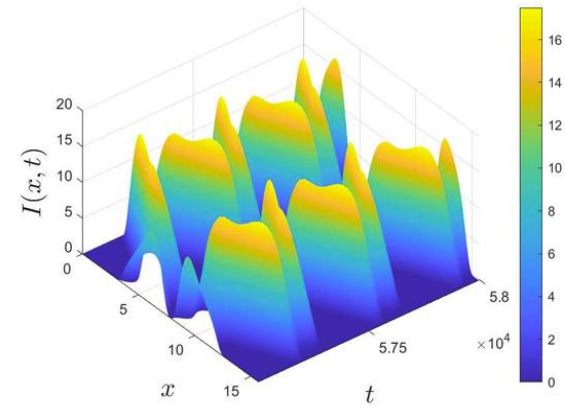


(b) Surface $I(x, t)$

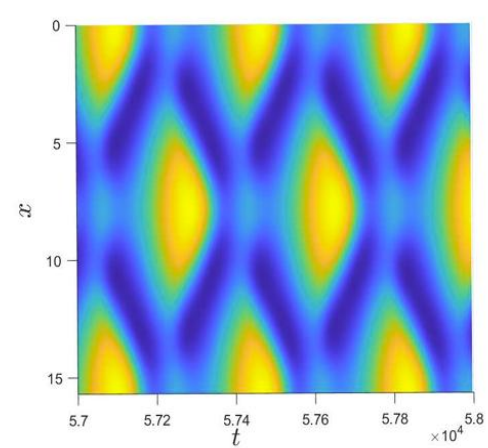
(k_1, k_2) – Turing Hopf Bifurcation



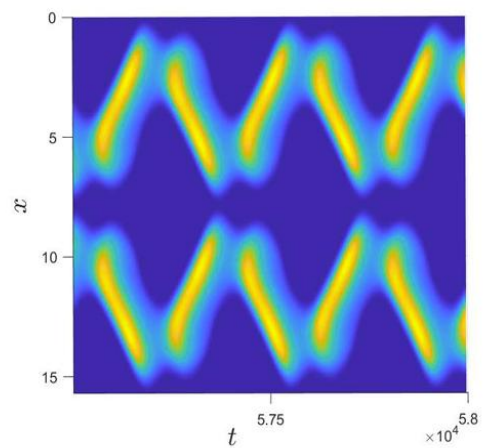
(a) Surface $S(x, t)$



(b) Surface $I(I, t)$

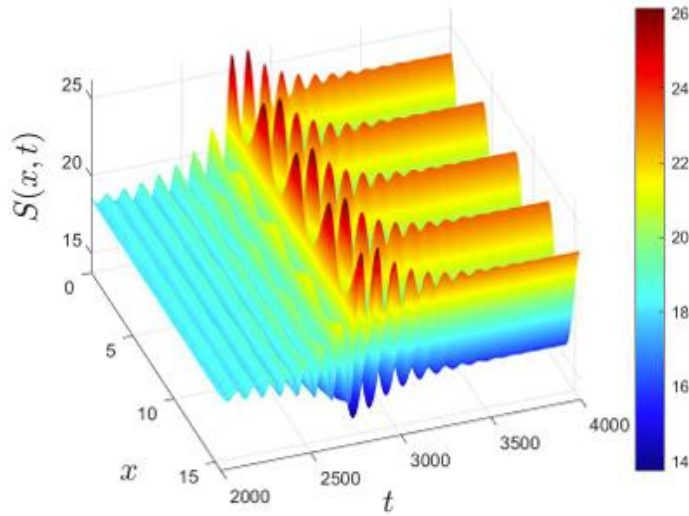


(a) Projection of $S(x, t)$

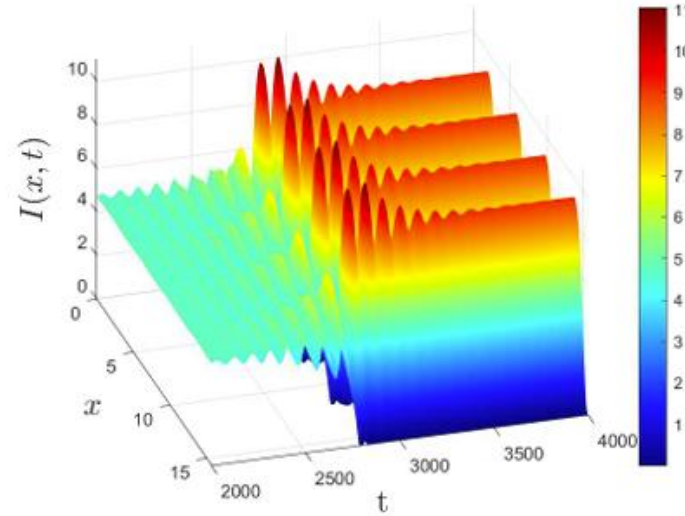


(b) Projection of $I(x, t)$

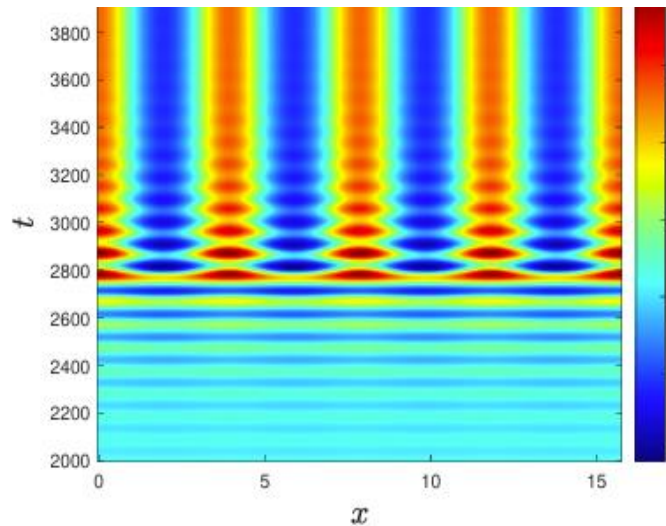
Transient dynamics from temporal oscillatory regime to a spatially periodic pattern



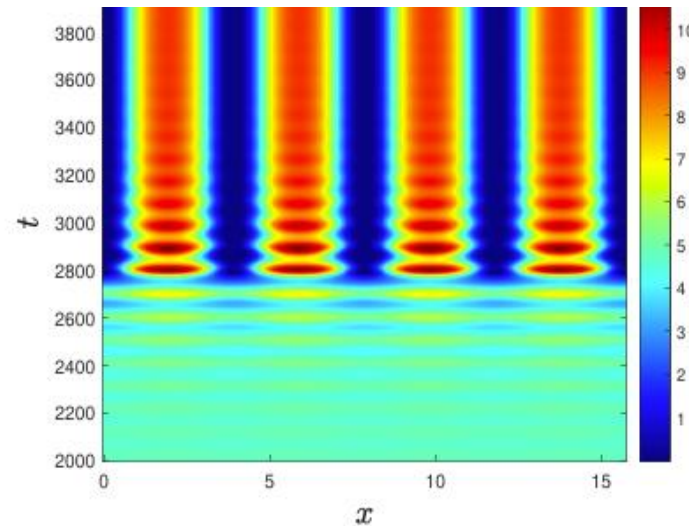
(a) Surface of $S(x, t)$



(b) Surface of $I(x, t)$



(a) Projection of $S(x, t)$



(b) Projection of $I(x, t)$

Transient dynamics observed:

- Early stage ($t < 2800$): temporally periodic, spatially homogeneous oscillations.
- Post-transition ($t > 2800$): shift to spatiotemporal patterns with localized hotspots and wave structures.

Public Health implication:

- **Uniform interventions are insufficient:**

Strategies applied evenly across space (e.g., mass vaccination, broad policies) may not control outbreaks effectively.

- **Spatial targeting is essential:**

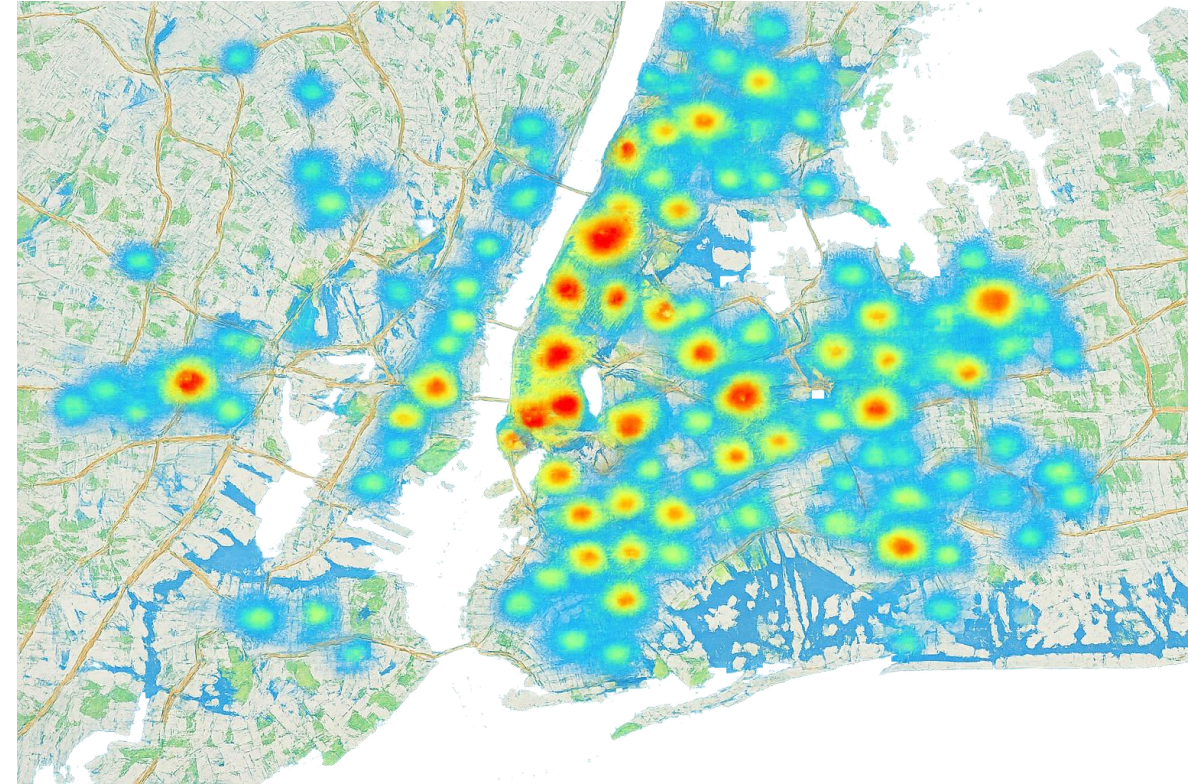
Interventions must adapt to **regional heterogeneity** and localized hotspots.

- **Cyclic and regional disease waves:**

Control policies should account for **asynchronous recurrence** of infection across different areas.

- **Adaptive strategy required:**

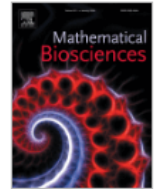
Combining **temporal timing** (when to intervene) with **spatial targeting** (where to intervene) is key to containment.





Mathematical Biosciences

Volume 392, February 2026, 109602



Original research article

Complex dynamics and pattern formation in a diffusive epidemic model with an infection-dependent recovery rate

Wael El Khateeb^a, Chanaka Kottegoda^b, Chunhua Shan^a  

- Anderson, R. M., & May, R. M. *Infectious Diseases of Humans: Dynamics and Control*. Oxford Univ. Press, 1991.
- Kermack, W. O., & McKendrick, A. G. “A contribution to the mathematical theory of epidemics.” *Proc. Royal Soc. London*, 115, 700–721 (1927).
- Murray, J. D. *Mathematical Biology II: Spatial Models and Biomedical Applications*. Springer, 3rd ed., 2003.
- Capasso, V., & Serio, G. “A generalization of the Kermack–McKendrick deterministic epidemic model.” *Math. Biosci.*, 42, 43–61 (1978).
- Gierer, A., & Meinhardt, H. “A theory of biological pattern formation.” *Kybernetik*, 12(1), 30–39 (1972).
- Jiang, W., An, Q., & Shi, J. “Formulation of the normal form of Turing–Hopf bifurcation in partial functional differential equations.” *J. Differ. Equ.*, 268, 6067–6102 (2020).
- Shan, C., & Zhu, H. “Bifurcations and complex dynamics of an SIR model with the impact of the number of hospital beds.” *J. Differ. Equ.*, 257, 1662–1688 (2014).

Thank you!

Wael El Khateeb
Ph.D. Candidate in Applied Math
The University of Toledo



LinkedIn



Personal Webpage